

# Chi-square test



There is a statistical test used to compare **observed results** with **expected results**. This Chi-Square Test helps to see if there's a significant difference between them.

## Types of Chi-Square Tests:

### 1. Chi-Square Test for Goodness of Fit

- Used to see if a sample matches a population.
- Example: Do dice roll results match the expected equal distribution?

### 2. Chi-Square Test for Independence

- Used to see if two variables are related or independent
- Example: Is there a relationship between gender and subject preference?

# Chi-square test



- Chi-square test should be applied only for frequencies. We cannot use them for data in terms of percentages, proportions, means or similar statistical contents.
- Chi-squared is generally applied for nominal rather than ordinal data. The data will be presented in an  $r \times c$  contingency table, where  $r$  is the number of rows and  $c$  is the number of columns. If  $r = 2$  and  $c = 2$ , then it is called  $2 \times 2$  contingency table. In Machine learning it is called as Confusion matrix.
- Chi-square test measures the relationship between two categorical variables, but as this relation may be based on chance, it is appropriate to call it as **ASSOCIATION** rather than RELATION

# Chi-square Test



It is very important to verify the Assumptions of Chi-square test

If the expected cell frequencies is **< 5**

→ **Yate's correction** should be applied for continuity

In a **2 x 2 contingency table**, if one or more of the cell has the expected cell frequencies is **< 5**,

→ Fisher's exact probabilities should be computed

For the use of **Chi-square test**

→ The sample size should not be **less than 20**.

The Fisher's exact Probability

$$P = \frac{1}{n!} \frac{r_1!}{a!} \frac{r_2!}{b!} \frac{c_1!}{c!} \frac{c_2!}{d!}$$

For an  $r \times c$  table, if the expected frequencies in any cells are **< 5**, merge the rows or columns meaningfully

# Chi-square Test



## 2 x 2 Contingency Table for testing independence

| Categorical Variable 1 | Categorical variable 2 |                | Total |
|------------------------|------------------------|----------------|-------|
|                        | Present                | Absent         |       |
| Present                | $O_1$<br>$E_1$         | $O_2$<br>$E_2$ | $r_1$ |
| Absent                 | $O_3$<br>$E_3$         | $O_4$<br>$E_4$ | $r_2$ |
| Total                  | $c_1$                  | $c_2$          | $n$   |

Calculation  
of expected  
frequencies

$$E_1 = \frac{r_1 c_1}{n}$$

$$E_2 = \frac{r_1 c_2}{n}$$

$$E_3 = \frac{r_2 c_1}{n}$$

$$E_4 = \frac{r_2 c_2}{n}$$

# Chi-square Test



## Hypothesis for testing independence

The hypothesis to be tested for independence will be

**$H_0$ : The two categorical variables may be independent  
(may not be associated)**

**Observed = Expected**

**$H_1$ : The two categorical variables may not be independent  
(may be associated)**

**Observed  $\neq$  Expected**

# Chi-square Test



## Procedure for testing independence

To check the independence (no association) between the two categorical variables, the statistical test used is Chi-square test given by

$$\chi^2 = \sum_{i=1}^m \sum_{j=1}^n \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

$$E_{ij} = \frac{r_i c_j}{n},$$

for  $i = 1, 2, \dots, m; j = 1, 2, \dots, n$

The test-statistic follows Chi-square distribution with  $(r - 1)(c - 1)$  degrees of freedom.  $r = \#$  of rows,  $c = \#$  of columns

# Example



An RTO department has observed that, on a highway, a greater number of accidents have taken place in the early hours of the day than other timings and wish to associate the outcome of the accidents with timings. A survey findings show that, of the 400 accident cases studied, 280 had met with accident in the early hours and 99 of them were fatal. Further, those who met with accident in the early hours and died was 80. Does this data indicate any association between the time of accident and fatality of the accident. Use  $\alpha = 0.05$ .

**Solution:**

**Hypothesis:**

- **$H_0$ : Timings of accident and its outcome are independent (Not associated)**
- **$H_1$ : Timings of accident and its outcome are dependent (Associated)**

# Solution



| Early hours of accident | Outcome of accident |           | Total |
|-------------------------|---------------------|-----------|-------|
|                         | Fatal               | Non-fatal |       |
| Yes                     | 80                  | 200       | 280   |
| No                      | 19                  | 101       | 120   |
| Total                   | 99                  | 301       | 400   |

$$E_1 = \frac{r_1 c_1}{n} = \frac{280 * 99}{400} = 69.3$$

$$E_3 = \frac{r_2 c_1}{n} = \frac{120 * 99}{400} = 29.7$$

$$E_2 = \frac{r_1 c_2}{n} = \frac{280 * 301}{400} = 210.7$$

$$E_4 = \frac{r_2 c_2}{n} = \frac{120 * 301}{400} = 90.3$$

# Solution



## Calculation of Chi-square statistic - $\chi^2$

| Sl No | Observed frequencies ( $O_i$ ) | Expected frequencies ( $E_i$ ) | $(O_i - E_i)$        | $(O_i - E_i)^2$ | $\frac{(O_i - E_i)^2}{E_i}$ |
|-------|--------------------------------|--------------------------------|----------------------|-----------------|-----------------------------|
| 1     | 80                             | 69.3                           | 10.7                 | 114.49          | 1.652                       |
| 2     | 200                            | 210.7                          | -10.7                | 114.49          | 0.543                       |
| 3     | 19                             | 29.7                           | -10.7                | 114.49          | 3.855                       |
| 4     | 101                            | 90.3                           | 10.7                 | 114.49          | 1.268                       |
| Total | 400                            | 400                            | Chi-square statistic |                 | $\chi^2 = 7.318$            |

# Chi-square Table



## Testing of Hypothesis → Chi-square test: Independence

Percentage Points of the Chi-Square Distribution

| Degrees of Freedom | Probability of a larger value of $\chi^2$ |       |       |       |       |       |       |       |       |
|--------------------|-------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|
|                    | 0.99                                      | 0.95  | 0.90  | 0.75  | 0.50  | 0.25  | 0.10  | 0.05  | 0.01  |
| 1                  | 0.000                                     | 0.004 | 0.016 | 0.102 | 0.455 | 1.32  | 2.71  | 3.84  | 6.63  |
| 2                  | 0.020                                     | 0.103 | 0.211 | 0.575 | 1.386 | 2.77  | 4.61  | 5.99  | 9.21  |
| 3                  | 0.115                                     | 0.352 | 0.584 | 1.212 | 2.366 | 4.11  | 6.25  | 7.81  | 11.34 |
| 4                  | 0.297                                     | 0.711 | 1.064 | 1.923 | 3.357 | 5.39  | 7.78  | 9.49  | 13.28 |
| 5                  | 0.554                                     | 1.145 | 1.610 | 2.675 | 4.351 | 6.63  | 9.24  | 11.07 | 15.09 |
| 6                  | 0.872                                     | 1.635 | 2.204 | 3.455 | 5.348 | 7.84  | 10.64 | 12.59 | 16.81 |
| 7                  | 1.239                                     | 2.167 | 2.833 | 4.255 | 6.346 | 9.04  | 12.02 | 14.07 | 18.48 |
| 8                  | 1.647                                     | 2.733 | 3.490 | 5.071 | 7.344 | 10.22 | 13.36 | 15.51 | 20.09 |
| 9                  | 2.088                                     | 3.325 | 4.168 | 5.899 | 8.343 | 11.39 | 14.68 | 16.92 | 21.67 |
| 10                 | 2.558                                     | 3.940 | 4.865 | 6.737 | 9.342 | 12.55 | 15.99 | 18.31 | 23.21 |

# Conclusion



## Hypothesis

**$H_0$ :** Timings of accident and its outcome are independent  
(Not associated)

**$H_1$ :** Timings of accident and its outcome are dependent  
(Associated)

$$\chi^2 = 7.31$$

**$df = 1$ ,** Critical value at  $\alpha = 0.05$  is 3.841,

**Inference:** There may be an association between timings of accident and its outcome

# Example



In an experiment to study the dependence of hypertension on smoking habits, the following data were taken on 180 individuals

| Hypertension | Non smokers | Moderate smokers | Heavy smokers | Total |
|--------------|-------------|------------------|---------------|-------|
| Yes          | 21          | 36               | 30            | 87    |
| No           | 48          | 26               | 19            | 93    |
| Total        | 69          | 62               | 49            | 180   |

# Solution



## • Hypothesis:

**$H_0$ : Smoking habit and hypertension may be independent (may not be associated)**

**$H_1$ : Smoking habit and hypertension may not be independent (may be associated)**

**Calculation of expected frequencies**

$$E_1 = \frac{(r_1 \times c_1)}{n} = (87 \times 69)/180 = 33.35$$

$$E_2 = \frac{(r_1 \times c_2)}{n} = (87 \times 62)/180 = 29.97$$

$$E_3 = \frac{(r_1 \times c_3)}{n} = (87 \times 49)/180 = 23.68$$

$$E_4 = \frac{(r_2 \times c_1)}{n} = (93 \times 69)/180 = 35.65$$

$$E_5 = \frac{(r_2 \times c_2)}{n} = (93 \times 62)/180 = 32.03$$

$$E_6 = (r_2 \times c_3)/n = (93 \times 49)/180 = 25.32$$

# Solution



## Calculation of Chi-square statistic

| Sl No | (O <sub>i</sub> ) | (E <sub>i</sub> ) | (O <sub>i</sub> - E <sub>i</sub> ) | (O <sub>i</sub> - E <sub>i</sub> ) <sup>2</sup> | $\frac{(O_i - E_i)^2}{E_i}$ |
|-------|-------------------|-------------------|------------------------------------|-------------------------------------------------|-----------------------------|
| 1     | 21                | 33.35             | - 12.35                            | 152.52                                          | 4.57                        |
| 2     | 36                | 29.97             | 6.03                               | 36.36                                           | 1.21                        |
| 3     | 30                | 23.68             | 6.32                               | 39.94                                           | 1.69                        |
| 4     | 48                | 35.65             | 12.35                              | 152.52                                          | 4.28                        |
| 5     | 26                | 32.03             | - 6.03                             | 36.36                                           | 1.14                        |
| 6     | 19                | 25.32             | - 6.32                             | 39.94                                           | 1.58                        |
| Total | 180               | 180               | Chi-square statistic               |                                                 | $\chi^2 = 14.46$            |

# Interpretation



## Interpretation

**$H_0$ :** Smoking habit and hypertension may be independent (may not be associated)

**$H_1$ :** Smoking habit and hypertension may not be independent (may be associated)

$$\chi^2 = 14.46$$

**$df = 2$ ,** Critical value at  $\alpha = 0.05$  is 5.99,  **$P < 0.001$**

**Inference:** There may be an association between smoking and Hypertension

# Chi-square test for Goodness-of-fit



A powerful test for testing the significance of the discrepancy between theory and experiment was given by Karl Pearson known as “Chi-square test for Goodness – of – fit”. It enables to find the deviation of the experiment from theory is just by chance or is it really due to the inadequacy of the theory to fit the observed data.

If  $O_i$  ( $i = 1, 2, \dots, n$ ) is a set of observed (experimental) Frequencies and  $E_i$  ( $i = 1, 2, \dots, n$ ) is the corresponding set of expected (theoretical or hypothetical) frequencies, then the Chi-square test statistic is given by

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} \approx \chi^2_{(n-1)}$$

follows Chi-square distribution with  $n - 1$  degree of freedom.

# Example:



The following data shows the distribution of digits in numbers chosen at random from a telephone directory. The digits are:

| Digits | 0    | 1    | 2   | 3   | 4    | 5   | 6    | 7   | 8   | 9   | Total |
|--------|------|------|-----|-----|------|-----|------|-----|-----|-----|-------|
| f      | 1026 | 1107 | 997 | 966 | 1075 | 933 | 1107 | 972 | 954 | 853 | 10000 |

$H_0$ : The digits occur uniformly frequently in the directory

$H_1$ : The digits do not occur uniformly frequently in the directory

# Solution



Under the null hypothesis, the expected frequency for each of the digits 0, 1, ..., 9 is  $10000 \div 10 = 1000$ . The Chi-square value is

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i} = \frac{(1026 - 1000)^2}{1000} + \dots + \frac{(853 - 1000)^2}{1000}$$
$$\chi^2 = 58.542$$

Since  $\chi^2 = 58.542 < 16.919$  (critical value,  $df = 9$ ), it can be infer that the digits are not uniformly distributed.

# Example 2



A consultant was employed by a city council to study the pattern of bus arrival and departure at a very busy interstate bus terminus. She collected data from the arrival of 200 buses. Based on the data, the average arrival time was found to be  $\lambda = 2.96$ . She divided the arrivals into 6 categories. Assuming that the arrivals follow Poisson distribution test whether the arrival distribution follows Poisson law. Use  $\alpha = 0.01$ .

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**Solution:** The probabilities are to be calculated using Poisson distribution with  $\lambda = 2.96$  and  $x = 0, 1, 2, 3, 4$ , and  $\geq 5$ . The results are as follows:

# Solution



The probabilities are to be calculated using Poisson distribution with  $\lambda = 2.96$  and  $x = 0, 1, 2, 3, 4$ , and  $\geq 5$ . The results are as follows:

| No. of arrivals | $O_i$ | $p_i$  | $E_i$ | $\frac{(O_i - E_i)^2}{E_i}$ |
|-----------------|-------|--------|-------|-----------------------------|
| 0               | 10    | 0.0524 | 10.48 | 0.0220                      |
| 1               | 13    | 0.1545 | 30.90 | 10.3693                     |
| 2               | 45    | 0.2277 | 45.54 | 0.0064                      |
| 3               | 49    | 0.2238 | 44.76 | 0.4016                      |
| 4               | 32    | 0.1651 | 33.02 | 0.0315                      |
| $\geq 5$        | 41    | 0.1765 | 35.30 | 0.9204                      |
| $\chi^2 =$      |       |        |       | 11.7512                     |

Since  $\chi^2 = 3.402 < 13.27$  (critical value at  $df = k-2 = 4$ ), it can be inferred that the arrivals and departures follow Poisson law.

# Exercise:



1. Aircrew escape system are powered by a solid propellant. The burning rate of this propellant is an important product characteristic. Specification require that the mean burning rate must be 50 cm/s. From past experience it is known that the population SD is 2 cm/s. A sample of 25 solid propellant were selected randomly to reconfirm the specification stated. The sample mean found was 51.3 cm/s. At 5% level of significance what conclusion should be drawn regarding burning rate?

**Hint:**  $\mu=50$ ,  $\sigma=2$ ,  $n=25$ ,  $\bar{X}=51.3$ ,  $H_0: \mu=50$ ,  $H_1: \mu \neq 50$ , (Two tailed test),  $Z = (\bar{X} - \mu) / (\sigma/\sqrt{n})$ ,  $Z_{\text{critical}} = \pm 1.96$

2. A builder claims that solar water heater are installed in 70% of all homes being constructed today in a city. Would you agree with this claim if a random sample of new homes in this city shows that 28 out of 55 had heat pumps installed? What P-value and confidence interval are related in this situation?

**Hint:**  $H_0: p=0.7$

$H_1: p \neq 0.7$ .

$SE = \sqrt{pq/n}$ , choose 95% confidence interval for a two tailed test.

$$CI = \hat{p} \pm Z \times SE$$

# Exercise



3. A cigarette manufacturing company claims that its brand A cigarettes outsells its brand B cigarettes by 8%. If it is found that 42 out of a random sample of 200 smokers prefer brand A and 18 out of 100 smokers prefer brand B, test at 5% level of significance, whether 8% difference a valid claim. Also construct 95% CI for  $(P_1 - P_2)$  and find P-value.

**Hint:**  $n_1=200$ ,  $n_2=100$ ,  $x_1=42$ ,  $x_2=18$ ,

Null hypothesis:  $p_1 - p_2 = 0.08$ , alternative hypothesis:  $p_1 - p_2 \neq 0.08$ ,

Find pooled SE where,  $p=0.2$  and  $q=0.8$ ,  $CI=0.03 \pm 0.096$

4. The management of a local health club claims that its members lose on the average 7 kgs or more within 3 months after joining the club. To check this claim, a consumer agency took a random sample of 15 members of this health club and found that they lost an average of 6.26 kgs within the first three months of membership. The sample standard deviation 1.91 kgs. Test at 1% level of significance whether the claim made by management of a local health club is acceptable or not? Also find the P-value of this test.

**Hint:**  $\alpha=0.01$ , one tailed

$H_0: \mu \geq 7$ ,  $H_1: \mu < 7$ ,  $n=15$  ( $<30 \Rightarrow$  t-test),  $df=15-1=14$ ,  $\bar{x} = 6.26$ ,  $s=1.91$ ,  $p\text{-value} = 0.075$  (approx value via t table between 0.05 to 0.10 at  $|t|=1.5$ ,  $df=14$ )

# Exercise



5. A taxi company manager is trying to decide whether the use of radial tires instead of regular belted tires improves fuel economy. 12 cars were equipped with radial tires and driven over a prescribed test course. Without changing drivers, the same cars were then equipped with regular belted tires and driven once again over the test course. The gasoline consumption in kilometers per liter was recorded. Assume that the populations are normally distributed

|              |     |     |     |     |     |     |     |     |     |     |     |     |
|--------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Radial tyres | 4.2 | 5.7 | 2.6 | 7.0 | 6.7 | 4.5 | 5.7 | 5.1 | 7.4 | 6.9 | 6.1 | 3.2 |
| Belted tyres | 4.3 | 3.9 | 5.2 | 5.9 | 5.8 | 4.4 | 4.7 | 5.8 | 5.9 | 3.7 | 4.9 | 4.9 |

- What are the underlying assumptions of Student's unpaired t-test?
- Based on the data can we conclude that cars equipped with radial tires give better fuel economy than those equipped with belted tires?
- Find P-value.
- Construct 95% CI for difference in means.
- What is your inference?

# Hints for Q5



$H_0$ : There is no difference in fuel economy between radial and regular belted tires.

$H_1$ : Radial tires give better fuel economy than belted tires.

$df=11$ ,

One tailed test with 95% CI

$\Rightarrow \alpha=0.05$  and  $t(0.05, 11)=1.796$

Calculate  $t$  using student's unpaired  $t$ -test and compare with 1.796 to accept or reject the null hypothesis.

# Exercise



6. Three different analytical tests can be used to determine the impurity level in steel alloys. Each specimen is tested using two procedures and the results are shown in the following tabulation. Is there sufficient evidence to conclude that the three tests give the same mean impurity level using  $\alpha = 0.01$

| Specimen | 1   | 2   | 3   | 4   | 5   | 6   | 7   | 8   |
|----------|-----|-----|-----|-----|-----|-----|-----|-----|
| Test 1   | 1.2 | 1.3 | 1.5 | 1.4 | 1.7 | 1.3 | 1.6 |     |
| Test 2   | 1.4 | 1.7 | 1.5 | 1.3 | 2.0 | 2.1 | 1.7 | 1.9 |

**Hint:**  $H_0$ : Mean impurity level for all 3 tests are same.

$H_1$ : Mean impurity level for atleast one test is different.

Find mean and variance for both data, and compute the test statistic (t-test), consider  $df = (15-1) + (6-1) = 19$ .

# Exercise



7. Random samples of 15 and 10 were selected from two thermocouples. The sample means were 315, 303 and sample standard deviations were 3.8, 4.9 respectively.

- (a) Construct 95% CI for difference in means
- (b) Test whether there is any significant difference in the means of two thermocouples at 5% level of significance
- (c) Find the P-value

8. Jaswant is interested in breeding flowers of a certain species. The experimental breeding can result in four possible types of flowers

- (a) Megenta flowers with green stigma (MG)
- (b) Megenta flowers with red stigma (MR)
- (c) Red flowers with green stigma (RG)
- (d) Red flowers with red stigma (RR)

According to Mendel's law, these four kinds of flowers should come out in the ratio of 9 : 3 : 3 : 1. Jaswant found that under her experiment, out of 160 flowers that bloomed the number of flowers with types MG, MR, RG, and RR were 84, 35, 28, and 13. She wants to find out, whether these data are compatible with Mendel's law. Use  $\alpha = 0.05$ .

# Exercise 9



## Diet-modification Program



Ten individuals have participated

| Subject       | 1   | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   | 10  |
|---------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Weight Before | 195 | 213 | 247 | 201 | 187 | 210 | 215 | 246 | 294 | 310 |
| Weight After  | 187 | 195 | 221 | 190 | 175 | 197 | 199 | 221 | 278 | 285 |

Is there sufficient evidence to support claim that this program is effective in reducing weight?

Assume that the difference between before and after follow normal distribution. Use  $\alpha = 0.05$ .

# Exercise 10



To assess the length of hospital stay and the type of insurance, data were taken on 70 individuals



| Type of Insurance | Length of Hospital Stay (days) |        | Total |
|-------------------|--------------------------------|--------|-------|
|                   | $\leq 10$                      | $> 10$ |       |
| Type 1            | 42                             | 3      | 45    |
| Type 2            | 18                             | 7      | 25    |
| Total             | 60                             | 10     | 70    |

Examine whether Chi-square test can be applied to this data to test the independence between type of insurance and length of hospital stay?

# Exercise 11



Three pension plans

Independent of job classification

Use  $\alpha = 0.05$

Examine whether there is any association between job classification and pension plan

The opinion of a random sample of 500 employees are shown below

| Job Classification | Pension Plan |     |     | Total |
|--------------------|--------------|-----|-----|-------|
|                    | 1            | 2   | 3   |       |
| Salaried workers   | 166          | 86  | 68  | 320   |
| Hourly workers     | 84           | 64  | 32  | 180   |
| Total              | 250          | 150 | 100 | 500   |